

Chern++: computational results on positivity for Morin Thom polynomials

Abstract

We report on a machine-checked study of the two positivity conjectures for Morin singularities. The infrastructure computes $\mathcal{Q}_d$ for $d \leq 6$ from the explicit orbit equations of Bérczi–Szenes §7.3, self-checked against every invariant fixed in advance. On top of it: Rimányi’s conjecture is verified for A_6 (first time), the two reductions of the A_5 handoff note are shown to survive at $d = 6$ while the proposed prefix-positivity route is shown to *fail* at $d = 6$, and a notion of *denominator certificate* is introduced whose successes are proofs and whose failures are theorems. The certificate machinery reproduces the $d = 4$ strong positivity theorem and Proposition 3 of the handoff note mechanically. It does not close the A_6 tail, and we show this is a limitation of the technique rather than of the search: after generalising Lemma 1 to an absorption criterion, 13 090 candidate certificates fail. Notation follows `rimanyi_positivity.pdf`, `report.pdf` and `a5_weak_positivity_handoff.pdf`, cited as [N], [R], [H].

1 Setting and notation

For the Morin singularity A_d , the Bérczi–Szenes formula expresses Tp_{A_d} as an iterated residue whose only d -dependent input is $\mathcal{Q}_d = \mathrm{mdeg}_{T_d}(\mathcal{O}_d)$, the equivariant multidegree of the Borel orbit closure $\mathcal{O}_d = \overline{B_d \cdot \epsilon_{\mathrm{ref}}}$ inside $\widehat{N}_d \subset \mathrm{Hom}(\mathbb{C}^d, \mathrm{Sym}^2 \mathbb{C}^d)$. Normalising $z_d = 1$ and setting $x_j = z_j/z_{j+1}$, positivity is read off the chamber series

$$F_d(x) = \frac{\prod_{m < l} (1 - z_m/z_l) \mathcal{Q}_d}{\prod_{m+r \leq l} (1 - z_m/z_l - z_r/z_l)} = \frac{N_d(x)}{\prod_r (1 - f_r(x))} = \sum_{\beta \geq 0} A_\beta x^\beta,$$

where every f_r has nonnegative coefficients and zero constant term. Write $\tau(i, j, k, \dots) = (j - i, j, k, \dots)$ for the first adjacent transposition, and $C(M)$ for the ℓ -free Chern coefficient of [R, Thm. 5].

2 Infrastructure

2.1 Computing $\mathcal{Q}_d$

We take the route of [N, §7.3]: rather than parametrise the orbit and eliminate the group variables, we write down the quadratic relations of Proposition 7.3 directly, saturate by the defect-zero coordinates to remove the spurious components that appear from $d = 6$ on ([N, §7.4]), and read the multidegree off an initial ideal. Runtime for $d \leq 6$ is seconds.

The design point worth stating is that every quantity the computation can be checked against is known in advance, and all of them are checked: multihomogeneity of each relation under the torus weights; the codimension $\dim \widehat{N}_d - \binom{d}{2}$ forced by homogeneity of the residue formula; vanishing on random B_d -translates of ϵ_{ref} ; $\deg \mathcal{Q}_d$; exactness of the chamber division; and the normalisation fixing the c_1^d coefficient. The stage refuses to emit an artifact otherwise. Externally, all eleven classical A_6 coefficients are reproduced.

2.2 Scale of the objects

Table 1 is the practical reason sampling is no substitute for a proof: roughly half the coefficients of N_d are negative at every order, and the counts grow fast.

Table 1: The mined chamber algebras. $\deg \mathcal{Q}_d = \dim \widehat{N}_d - \binom{d}{2}$ is the codimension of the orbit closure.

	vars	$\dim \widehat{N}_d$	$\deg \mathcal{Q}_d$	$\mathcal{Q}_d$ terms	N_d terms	$\deg N_d$	N_d negative
A_4	3	7	1	3	54	13	27 (50%)
A_5	4	13	3	19	814	31	414 (51%)
A_6	5	22	7	418	26618	66	13309 (50%)

2.3 Exactness

No result below depends on floating point. One point deserves emphasis because it is a trap: Laurent coefficients are accumulated in `int64`, and they outgrow it. A wrapped accumulator reads as a large *negative* number, that is, as a spurious counterexample to the very conjecture under test. The evaluator therefore replays the fixed point in floating point as a magnitude probe and raises rather than returning wrapped values; this is what caps the A_6 sweep at $\ell = 5$ below.

3 Results for A_6

Proposition 1 (verified). *Rimányi’s conjecture holds for A_6 at every relative dimension $\ell \leq 5$, and for A_5 at $\ell \leq 8$.*

These are finite exact computations, not proofs of the infinite statement. For A_5 , [R, Thm. 8] reaches $\ell \leq 11$ using unbounded-precision arithmetic; our contribution at $d = 5$ is speed rather than range, and the ℓ -free reduction makes the two nested anyway. The A_6 verification is new. Passing $\ell = 5$ requires exact arithmetic beyond `int64`; residue arithmetic modulo several primes with CRT reconstruction is the natural route.

The negative Laurent coefficients of F_6 behave much as at $d = 5$; see Table 2. The first counterexample to strong Laurent positivity occurs at $d = 6$ at the same β as at $d = 5$, extended by a zero. The last column is the one that matters in §3.2.

Table 2: Negative Laurent coefficients of the chamber series, to total degree 11.

	terms	negative	first negative	most negative	with $i = 0$
A_4	237	0	—	—	0
A_5	490	6	(1, 1, 2, 1)	−1	0
A_6	796	10	(1, 1, 2, 1, 0)	−2	2

3.1 The reductions of $[H]$ at $d = 6$

Observation 2. *Both reductions of $[H]$ survive at $d = 6$ throughout the tested range ($\deg \leq 11$): the unpaired tail $i > j \Rightarrow A_{i,j,\dots} \geq 0$, and the paired inequality $A_{i,j,\dots} + A_{j-i,j,\dots} \geq 0$ for $0 \leq i \leq j$.*

This is evidence that the τ -pairing frame is structural rather than an A_5 accident.

3.2 Prefix positivity fails at $d = 6$

[H, §8] observes that every tested coefficient of $B = F_5/(1 - a)$ is nonnegative, and [H, §11.4] proposes proving that as a stepping stone, since $B_{i,j,k,l} = \sum_{r \leq i} A_{r,j,k,l}$ turns the paired inequality into a statement about adjacent differences of a prefix array.

Proposition 3. $F_6/(1 - a)$ is not coefficientwise nonnegative. Indeed $A_{(0,2,3,2,2)} = -1$.

Proof. The prefix sum runs over $r \leq i$, so $B_{0,j,k,l} = A_{0,j,k,l}$ and any negative coefficient with $i = 0$ survives untouched. F_5 has no negative coefficient with $i = 0$ in the tested range; F_6 has two, the first being $A_{(0,2,3,2,2)} = -1$. Confirmed along two independent code paths, truncated series products in exact integers and the XLA fixed-point grid. $\square$

The route of [H, §11.4] is therefore specific to $d = 5$. Note that the obstruction is at the very first level of the prefix sum, so no refinement of the argument recovers it.

4 Denominator certificates

Definition 4. Let $N/\prod_{r=1}^R(1 - f_r)$ with each f_r nonnegative and of zero constant term. An *order- k certificate* is an identity

$$N = \sum_{|S| \leq k} P_S \prod_{r \in S} (1 - f_r), \quad \text{every } P_S \text{ coefficientwise nonnegative,}$$

over subsets $S \subseteq \{1, \dots, R\}$.

Dividing through, $F = \sum_S P_S / \prod_{r \notin S} (1 - f_r)$ is a sum of products of nonnegative series, so $F \geq 0$. Order 0 is the statement that N itself is nonnegative. [H, §10.3] reports a first-order search of this shape coming back infeasible.

4.1 Why a returned certificate is a proof

$P_\emptyset$ is not a free unknown: since $\prod_{r \in \emptyset} (1 - f_r) = 1$, it is the remainder $N - \sum_{S \neq \emptyset} P_S \prod_{r \in S} (1 - f_r)$. We therefore solve a feasibility LP in the nonempty parts only, snap the solver's floating-point answer to exact rationals, and recompute $P_\emptyset$ in exact arithmetic. The identity then holds by construction, and the single remaining question, whether that exact remainder is coefficientwise nonnegative, is decided exactly. The LP is demoted to a search heuristic; correctness never depends on it. The output is a finite algebraic identity checkable in any computer algebra system.

4.2 Why a failure is a theorem

Lemma 5 (projection). *A constraint on a monomial of total degree T involves only the coefficients of the P_S of degree $\leq T$, because every $\prod_{r \in S} (1 - f_r)$ has constant term 1 and no negative-degree terms. Truncating both the constraints and the unknowns at degree T is therefore an exact projection of the feasible set, not a further restriction.*

Consequently infeasibility of the truncated program proves that no certificate of that order exists at *any* degree. Failed searches become impossibility results.

4.3 A_4

Proposition 6. F_4 admits an order-4 certificate with $\deg P_S \leq 8$, exhibited explicitly and verified exactly; and no certificate of order ≤ 3 exists at any degree. The minimal order is 4.

This is an independent computational proof of [N, Thm. 1] by a method using no special feature of $d = 4$. The lower bound already follows by hand at depth 1: writing m_S for the constant term of P_S , matching the constant term of $N = 1 - a - b - c + \dots$ forces $\sum_S m_S \leq 1$, while matching the coefficients of a, b, c forces

$$\sum_{S \ni f_0} m_S \geq \frac{1}{2}, \quad \sum_{S \ni f_2} m_S \geq 1, \quad \sum_{S \ni f_5} m_S \geq 1,$$

where $f_0 = 2a$, $f_2 = b + ab$, $f_5 = c + abc$ are the only factors containing the respective bare monomials. With a budget of 1 these can only be met if a single S contains all three. The machine-found certificate saturates all four bounds exactly, its only two constant terms being $\frac{1}{2}(1 - f_0)(1 - f_2)(1 - f_4)(1 - f_5)$ and $\frac{1}{2}(1 - f_2)(1 - f_4)(1 - f_5)(1 - f_6)$.

4.4 A_5 : how far the obstruction reaches

Proposition 7. No denominator certificate of order ≤ 7 exists at any degree for F_5 , nor for $F_5/(1 - a)$.

For F_5 this is expected, since $F_5 \not\geq 0$, and serves as a control that the engine never manufactures a certificate for a false statement. For the prefix series, which *is* coefficientwise nonnegative as far as we compute, it is informative: it extends [H, §10.3] from order 1 to order 7. The bound sharpens with probe depth: depth 1 gives ≥ 4 , depth 2 gives ≥ 7 , depth 3 rules out everything up to 7.

5 The unpaired tail

5.1 One series

Set $H(a) = (1 - a)/(1 - 2a)$ and $J_d = F_d/H$. After H is removed, every occurrence of a in a numerator or denominator monomial is accompanied by b : the chamber monomials containing a are $z_1/z_l = ab \cdots$, and only $z_1/z_2 = a$ is bare, which is exactly the factor $1 - 2a$ that H removes. Hence $\deg_a [b^j c^k \cdots] J_d \leq j$, and convolving with $H(a) = 1 + \sum_{n \geq 1} 2^{n-1} a^n$ gives, for $i > j$,

$$A_{i,j,k,\dots} = 2^{i-1} [b^j c^k \cdots] J_d(1/2, b, c, \dots).$$

The entire unpaired tail, infinitely many coefficients, is therefore one positivity question about a single series in $d - 2$ variables. This is [H, Lemma 2 and Prop. 3] for $d = 5$, stated so that it applies verbatim for any d .

Our $J_5(1/2, b, c, e)$ agrees with [H, eq. (9)] coefficient for coefficient to degree 12, which is what validates the construction. $J_6(1/2, b, c, d, e)$ is coefficientwise nonnegative as far as we compute ($\deg \leq 10$).

5.2 Multiplicative certificates

The two successful proofs in the literature, [N, Thm. 1] and [H, Prop. 3], are *multiplicative*: products of ratios $(1 - u)/(1 - v)$, each nonnegative by Lemma 1 because $v - u \geq 0$. That shape is not a finite sum $\sum_S P_S \prod_{r \in S} (1 - f_r)$ with $P_S \geq 0$, so the additive certificates of §4 structurally cannot find them.

The diagnostic is decisive: the additive search is obstructed on $J_5(1/2, \cdot)$, a series whose nonnegativity *is* Proposition 3. When a method fails on a known theorem, the method is wrong.

Presenting the numerator in factored form, the Vandermonde part being already a factorisation, one can search for a matching of numerator factors to denominators satisfying the Lemma 1 condition, cancel whatever divides exactly, and ask whether what remains is nonnegative.

Proposition 8. *For $d = 5$ this closes: all nine numerator factors pair off, three denominators cancel exactly, and the leftover numerator is exactly 1. Proposition 3 is reconstructed mechanically.*

For $d = 6$ it does not. A maximum matching leaves

$$\frac{1 - cde - \frac{3}{2}bcde}{1 - 2cde},$$

whose expansion has the negative family $b(cde)^k$, and no available denominator v dominates $u = cde + \frac{3}{2}bcde$. Maximum matching is in fact greedy in the wrong way: it consumes denominators the remainder needs. Returning two ratios to the remainder leaves twelve Lemma-1 ratios times a remainder (207-term numerator over nine denominators) that is nonnegative as far as we check.

5.3 Generalising Lemma 1, and a negative result

Lemma 1 requires the numerator in the shape $1 - u$. The following drops that and lets several denominators cooperate.

Lemma 9 (absorption). *Write $N = N_+ - N_-$ with both parts coefficientwise nonnegative, and let S be a set of denominators with $1 - W = \prod_{v \in S} (1 - v)$. If $N_- \leq N_+ \cdot W$ coefficientwise, then $N / \prod_{\text{all}} (1 - v)$ is coefficientwise nonnegative.*

Proof. $N - N_+ \prod_S (1 - v) = N_+ W - N_- \geq 0$, so $N = N_+ \prod_S (1 - v) + P$ with $P \geq 0$ and

$$\frac{N}{\prod_{\text{all}} (1 - v)} = \frac{N_+}{\prod_{v \notin S} (1 - v)} + \frac{P}{\prod_{\text{all}} (1 - v)},$$

both nonnegative. □

Lemma 1 is the case $N = 1 - u$, $S = \{v\}$, where the condition reads $u \leq v$; the criterion is strictly stronger, since it applies to numerators not of that shape.

Observation 10. *Searching jointly over back-off subsets and absorbing subsets, 13 090 candidate criteria, closes nothing for A_6 .*

Since $J_6(1/2, \cdot)$ is nonnegative as far as we compute, a proof exists. The content of the observation is that it is not of this form: the technique that proves $d = 5$ does not extend, and the obstruction is structural rather than a matter of searching harder. Concretely, the negativity of the residual factor is compensated by the *other* ratios, so no local pairing argument can succeed.

6 Assessment

What the computations suggest about proof strategy:

1. The τ -pairing reduction is the right frame, both its inequalities surviving at $d = 6$, but the prefix-positivity stepping stone is not, and is unavailable from $d = 6$ on.
2. Additive, Positivstellensatz-shaped certificates are obstructed on *every* target tried, including provable ones. On present evidence they are not the tool for this problem, and we would not recommend further effort extending [H, §10.3] in that direction.

3. Multiplicative certificates are the tool that works, and they are now mechanised: they reproduce both published proofs from scratch. Their reach stops before A_6 .
4. The natural remaining targets are the paired series $\mathcal{P}$ of [H, eq. (12)], for which the construction is now cheap since $S = J_5(1/2, \cdot)$ is exactly the object above, the reflection is an index reflection, and [H, §7] supplies the denominator; and closed-form domination along the negative families of [R, §7.2], where each family settled is infinitely many cases.

Reproducibility

Everything above is reproducible from a clean checkout of Chern++. Statements are separated throughout into finite exact verifications and proofs; the certificate machinery and the closed-form family arguments are the only components producing the latter. The annotated walkthrough is `src/examples.ipynb`, executed with its outputs.